Code No: 136AW **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
CRYPTOGRAPHY AND NETWORK SECURITY
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Draw the model for Network Security.	[2]
b)	Give a brief note on Steganography.	[3]
c)	Differentiate between block cipher and Stream cipher.	[2]
d)	Discuss the concept of diffusion and confusion in block ciphers.	[3]
e)	What is a digital signature and why is it important in cryptography?	[2]
f)	What role do certificates play in the distribution of public keys?	[3]
g)	List and define the primary threats to web security.	[2]
h)	What enhancements does IEEE 802.11i bring to wireless LAN security?	[3]
i)	List the security services offered by Authentication Header.	[2]
j)	What is IKE, and what role does it play in IPsec implementation?	[3]

PART – B

(50 Marks)

2.a) Give the classification of attacks.
b) Discuss the significance of key range and key size in cryptography. How do these factors affect the security and performance of cryptographic systems? [5+5]

OR

3.a) Explain Polyalphabetic substitution cipher with an example.
b) Compare and contrast symmetric and asymmetric key cryptography. [5+5]

4. What are the different modes of operation for block ciphers? Explain each in detail.[10]

OR

5. Explain how the Diffie-Hellman key exchange works. Include a discussion of its security implications. [10]

6. In detail, explain the working secure hash algorithm (SHA-512). [10]

OR

7. Describe the working mechanism of Kerberos V4. How does it ensure secure authentication in a network? [10]

8. Explain the need and operation of SSL protocol. [10]

OR

9.a) Discuss strategies to enhance mobile device security in a corporate environment.

b) Analyze the role of digital certificates in HTTPS. How do they contribute to the security of web communications? [5+5]

10.a) Explain how ESP provides confidentiality and integrity protection for IP packets.

b) Mention the benefits of combining multiple security associations in IPsec communication. [5+5]

OR

11.a) Explain how confidentiality and integrity are provided to an email by PGP.

b) What is Single Sign-On (SSO), and how does it streamline authentication across multiple systems? [5+5]

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